Urban Green Spaces and Mental Well-Being: A Cross-Sectional Study in Mid-Sized Cities

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Abstract

Urban green spaces have been shown to benefit mental health. This study examines the relationship between access to parks and self-reported well-being in residents of three mid-sized Indian cities. We surveyed 1,240 adults and collected data on park proximity, visit frequency, and mental health scores. Results show that residents who visit green spaces at least twice a week report significantly higher well-being scores. Distance from the nearest park is negatively correlated with well-being. These findings support urban planning policies that prioritize green space access. Future work should examine causal mechanisms and long-term effects.

Keywords: urban green space, mental health, well-being, park access, urban planning

1. Introduction

Cities are growing fast. More people live in urban areas now than ever before. This trend is expected to continue. As cities expand, access to nature often shrinks. Residents in dense areas may have few parks nearby. This is a public health concern.

Mental health problems are common in cities. Stress, anxiety, and depression affect millions of urban residents. Researchers have long suspected that contact with nature can help. Parks, trees, and open water may reduce stress and improve mood. Several studies support this idea.

India is urbanizing rapidly. Its cities differ in size and layout. Some have large parks and tree-lined streets. Others are dense with little green cover. This variation makes India a useful setting for studying green space effects.

This study asks a simple question: do people who live near parks feel better? We also ask whether visiting parks makes a difference. We collected data from three mid-sized Indian cities to answer these questions.

Section 2 reviews relevant literature. Section 3 describes our methods. Section 4 presents results. Section 5 discusses findings and their implications.

2. Literature Review

The extant corpus of empirical and theoretical scholarship pertaining to the psychophysiological ramifications of proximate exposure to naturalistic environments has, over the course of the preceding four decades, converged upon a set of robust and methodologically well-substantiated propositions that collectively suggest a non-trivial, mechanistically plausible, and culturally transcendent ameliorative relationship between sustained or recurrent contact with biotic and semi-naturalistic urban landscapes and the subjective, hedonic, and eudaimonic dimensions of human psychological functioning, as operationalized across diverse instrumentation paradigms including but not limited to the Warwick-Edinburgh Mental Wellbeing Scale, the Perceived Stress Scale, the General Health Questionnaire, and various neuroimaging and psychophysiological correlates of affective valence and arousal.

Attention Restoration Theory, originally formulated by Rachel and Stephen Kaplan in their seminal 1989 monograph, postulates that the effortless, involuntary attentional engagement elicited by natural stimuli—characterized by the four theoretically interrelated properties of fascination, extent, compatibility, and being away—serves to replenish the finite and depletable cognitive resource designated as directed attention, thereby attenuating the psychobiological sequelae of sustained voluntary attentional effort that manifest phenomenologically as mental fatigue, irritability, impaired executive function, and diminished capacity for prosocial deliberation and impulse regulation, processes that are in turn implicated in a broad spectrum of psychopathological outcomes ranging from subclinical burnout to clinically diagnosable anxiety and mood disorders.

Complementarily, Stress Recovery Theory, as advanced by Ulrich and colleagues through a series of foundational psychophysiological investigations employing ambulatory cardiovascular monitoring and self-report affective indices, predicates its explanatory mechanism upon the phylogenetically ancient and evolutionarily conserved preferential affective orientation of Homo sapiens toward savanna-like environmental configurations—specifically, those exhibiting moderate vegetative complexity, visual permeability, and water feature salience—and argues that exposure to such configurations rapidly downregulates sympathetic autonomic nervous system activation via parasympathetically mediated vagal pathways, reducing circulating cortisol and catecholamine concentrations and restoring hedonic baseline within temporal intervals that may span as few as three to five minutes of passive visual exposure, a finding corroborated by convergent neuroimaging evidence implicating the subgenual prefrontal cortex and amygdala in nature-induced affective recalibration.

More recent transdisciplinary elaborations have sought to integrate these foundational theoretical frameworks with microbiome science, immunomodulatory research, and affective neuroscience, proposing that phytoncide inhalation during forest immersion, mycobiome-mediated immunostimulation through dermal and respiratory contact with biodiverse soil substrates, and attentional network reconfiguration induced by the fractal geometry characteristic of natural canopy structures collectively constitute a multimodal biopsychosocial pathway through which urban green space exposure exerts its well-documented salutary effects on anxious symptomatology, depressive ideation, and subjective vitality, and that the therapeutic valence of these mechanisms may be potentiated by physical activity, social interaction, and mindfulness-congruent attentional orientations that nature settings disproportionately afford relative to built environments.

Epidemiological and geospatial analyses, including large-scale nationally representative cohort investigations conducted by Richardson and colleagues in the United Kingdom and Gascon and collaborators within the European PHENOTYPE consortium, have demonstrated dose-response relationships between objectively quantified residential greenness—typically operationalized through satellite-derived Normalized Difference Vegetation Index values or administrative park proximity measures—and both self-reported and clinically adjudicated mental health outcomes, relationships that persist after comprehensive adjustment for socioeconomic position, baseline health status, urbanicity, and a range of potential confounders, lending conditional credibility to causal inference despite the inherent cross-sectional and observational limitations characterizing the preponderance of available evidence.

Within the South Asian and specifically Indian urban health literature, however, systematic inquiry into these associations remains comparatively nascent, methodologically heterogeneous, and insufficiently attentive to the culturally specific valences attached to particular forms of green infrastructure—sacred groves, temple gardens, riverbank corridors—that may not map neatly onto Western typologies of urban parkland and which may carry distinct psychosocial meanings and afford qualitatively different modes of restorative engagement for populations whose relationships with naturalistic environments are mediated by religious cosmologies, agrarian memory, and communal ritual practices that substantially diverge from those implicit in prevailing theoretical models, underscoring the necessity of culturally sensitive, locally validated research paradigms.

3. Methods

3.1 Study Design

We used a cross-sectional survey design. Data were collected between January and June 2023. Three cities were selected: Mysuru, Nashik, and Bhubaneswar. These cities were chosen for their range of green space availability and similar population sizes.

3.2 Participants

We recruited adults aged 18 and older. Participants were selected using stratified random sampling within each city. Strata were defined by neighborhood income level. A total of 1,240 participants completed the survey. Response rates ranged from 74% to 81% across cities.

3.3 Measures

Well-being was measured using the Warwick-Edinburgh Mental Wellbeing Scale (WEMWBS). This scale has 14 items rated from 1 to 5. Higher scores indicate better well-being. The scale has strong validity and reliability across diverse populations.

Park access was measured in two ways. First, we used GPS data to calculate the distance from each participant's home to the nearest public park. Second, we asked how often participants visit parks in a typical week.

We also collected data on age, sex, income, employment status, and self-rated physical health. These variables were used as covariates in regression analyses.

3.4 Analysis

We used descriptive statistics to summarize the sample. Linear regression was used to test the relationship between park access and well-being. All models included the full covariate set. We also ran stratified analyses by city and income level. Statistical significance was set at $p < 0.05$.

4. Results

4.1 Sample Characteristics

The mean age of participants was 36.4 years. Women made up 52% of the sample. Most participants were employed full-time (61%). About one-third reported low household income. Mean WEMWBS score was 48.3 out of a possible 70.

4.2 Park Access and Well-Being

Distance to the nearest park ranged from 40 meters to 4.2 kilometers. The mean distance was 820 meters. About 38% of participants lived within 500 meters of a park. Frequent visitors (two or more times per week) had higher mean WEMWBS scores than infrequent visitors. This difference was 4.7 points and was statistically significant ($p < 0.001$).

In regression models, distance to the nearest park was negatively associated with well-being. Each additional kilometer was associated with a 2.1-point decrease in WEMWBS score. This effect held after controlling for age, sex, income, and physical health.

Visit frequency was positively associated with well-being. Visiting a park twice a week or more was associated with a 3.9-point increase in WEMWBS score compared to visiting less than once a week (95% CI: 2.6–5.2).

4.3 Stratified Results

Results were broadly similar across the three cities. The association was strongest in Nashik and weakest in Bhubaneswar. Income did not significantly moderate the relationship. Effects were slightly larger for women than men, though this difference was not statistically significant.

5. Discussion

Our findings are consistent with prior research. People who live near parks and visit them often report better mental health. This holds true even after accounting for income and other factors. The results support the view that green spaces benefit urban residents.

The effect sizes are modest but meaningful at the population level. A 4-point increase in WEMWBS scores represents a real difference in daily well-being. Small improvements in park access could benefit large numbers of people.

Our study has several limitations. The cross-sectional design limits causal inference. We cannot rule out that healthier people choose to live near parks. Self-reported data may also introduce bias. Future studies should use longitudinal designs and objective health measures.

The findings have practical implications. City planners should consider mental health when designing green spaces. Policies that increase park access in underserved neighborhoods could reduce health inequalities. Programs that encourage park use may also be beneficial.

Future research should examine what types of green spaces matter most. Sacred or community spaces may have different effects than formal parks. Cultural context likely shapes how people experience and benefit from nature in cities.

6. Conclusion

Urban green spaces are linked to better mental health. Our study adds to a growing body of evidence from India and other developing countries. City planners and public health officials should take this evidence seriously. Investing in parks is an investment in mental health.

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